

MOBILE AND APPLICATION DEVELOPMENT

# Android OS Architecture



*Umar Bashir*

**Umar Bashir**

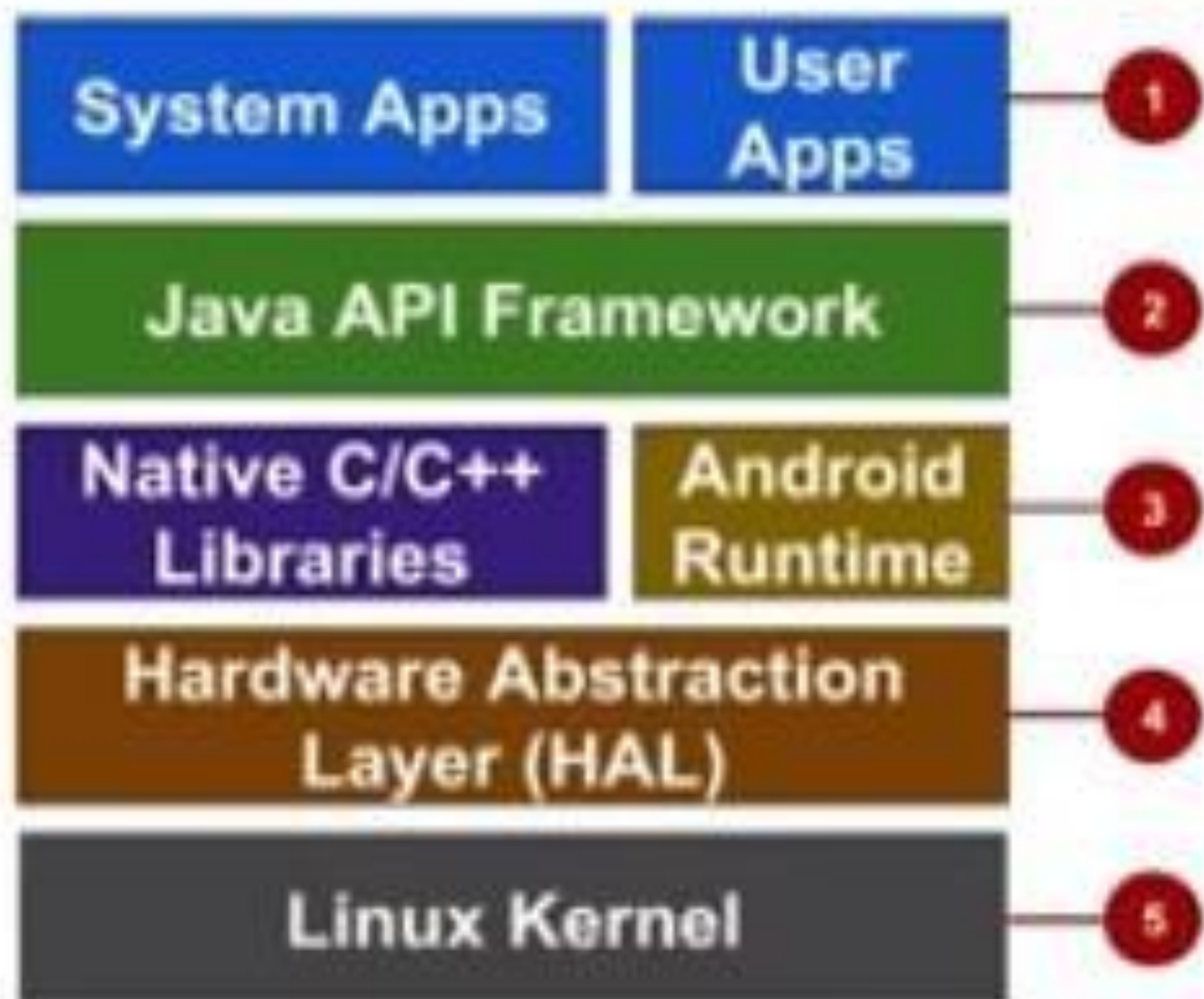
*Lecturer Computer Science*



android

# Android stack

1. System and user apps
2. Android OS API in Java framework
3. Expose native APIs; run apps
4. Expose device hardware capabilities
5. Linux Kernel



# System and user apps



- System apps have no special status
- System apps provide key capabilities to app developers

Example:

Your app can use a system app to deliver a SMS message.



# Java API Framework

- It contains all API's that we need for Android App Development and also called **SDK**.
- Persons who work in this Layer are called **SDK Developers**

The entire feature-set of the Android OS is available to you through APIs written in the Java language.

- View class hierarchy to create UI screens
- Notification manager
- Activity manager for life cycles and navigation
- Content providers to access data from other apps

# Android runtime

It is a place where all Applications come and run

Each app runs in its own process with its own instance of the Android Runtime.



<https://developer.android.com/>



After Android version 4, all devices are using ART

# C/C++ libraries

Image Processing Libraries  
like Open GL  
Like SSL

- It contains System Libraries

- Core C/C++ Libraries give access to core native Android system components and services.

- Developers who work in this Layer are called Android NDK Developers



# Hardware Abstraction Layer (HAL)

- It contains predefined methods for Hardware Communication
- Standard interfaces that expose device hardware capabilities as libraries

Examples: Camera, bluetooth module



# Linux Kernel

- Threading and low-level memory management
- Security features
- Drivers

